

# Problem Statement No. 22

## Interview Trainer Agent

**Program:** IBM SkillsBuild AICTE Internship on Emerging Technologies (AI & Cloud)

**Organized by:** Edunet Foundation, in partnership with AICTE and IBM SkillsBuild

**Student:** Deepanvitha Katari

**Project Name:** InterviewTwin – RAG-Powered Personalized Interview Trainer Agent

### The Challenge

An Interview Trainer Agent, powered by RAG (Retrieval-Augmented Generation), prepares users for job interviews by generating tailored question sets and preparation strategies based on their profile name, experience level, and job role.

It retrieves role-specific interview questions, industry expectations, behavioral scenarios, and HR guidelines from recruitment portals, professional networks, and company interview databases.

Users can input their resume or job title, and the agent provides targeted questions, model answers, and improvement tips.

It supports both technical and soft skill assessment, ensuring a comprehensive interview prep experience.

This AI-driven assistant builds user confidence, sharpens responses, and increases success rates in competitive hiring environments.

### Mandatory Technology

Use of IBM Cloud Lite services / IBM Granite is mandatory.

### Solution Implemented: InterviewTwin

InterviewTwin implements this problem statement as a full RAG pipeline: resume and job description analysis via IBM Granite, a gap analysis comparing candidate skills against job requirements, semantic retrieval over a 1,400+ question knowledge base (technical, behavioral, and HR questions across 8 roles) embedded using IBM Granite's embedding model, and a personalized, adaptive mock interview where question difficulty adjusts based on answer quality. Each answer is scored and evaluated by Granite, and the candidate receives a final performance report, the option to retry their weakest answer, and a personalized 7-day preparation plan — all grounded in IBM Cloud Lite's watsonx.ai and IBM Granite models, built using IBM Bob as the development environment.

GitHub Repository: <https://github.com/deepanvithakatari/interview-trainer-agent>